Freelance Marketplace Platform — Functionality, Services & Security Overview :

Executive Summary

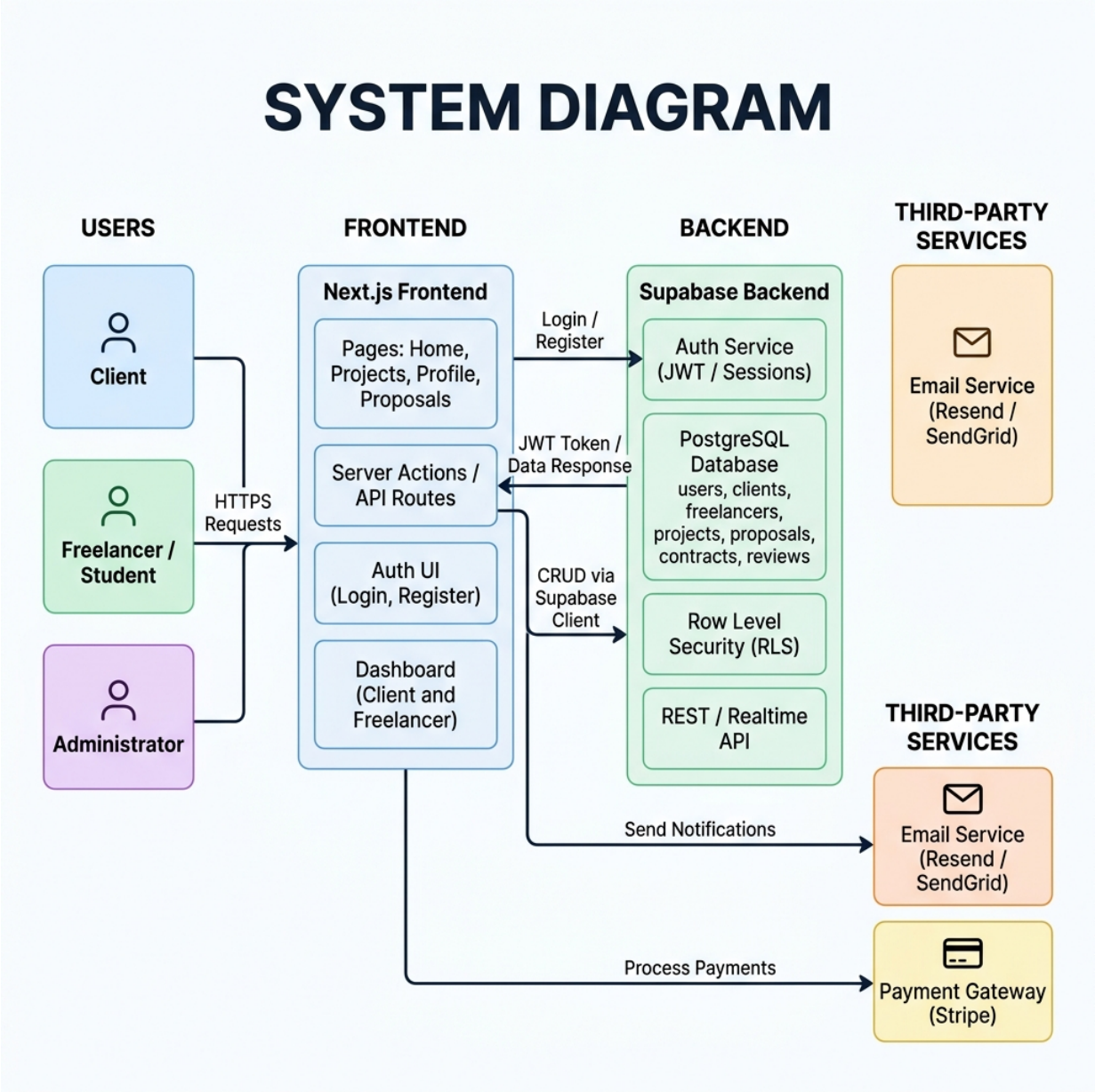
This platform is a full-stack freelance marketplace that connects Clients who need skilled work done with Freelancers who can deliver it. It is built on a modern, scalable technology stack — Next.js on the frontend and Supabase (PostgreSQL) on the backend — and integrates third-party services for payments (Stripe) and email notifications (Resend / SendGrid).

This report consolidates the full system architecture, all platform services, database design, user roles and privileges, and a comprehensive security analysis derived from the platform specification and supporting diagrams.

System Architecture Overview

2.1. System Diagram

The system follows a four-tier architecture: Users → Next.js Frontend → Supabase Backend → Third-Party Services. All client–server communication is encrypted via HTTPS. Authentication is JWT-based, managed entirely by Supabase Auth.



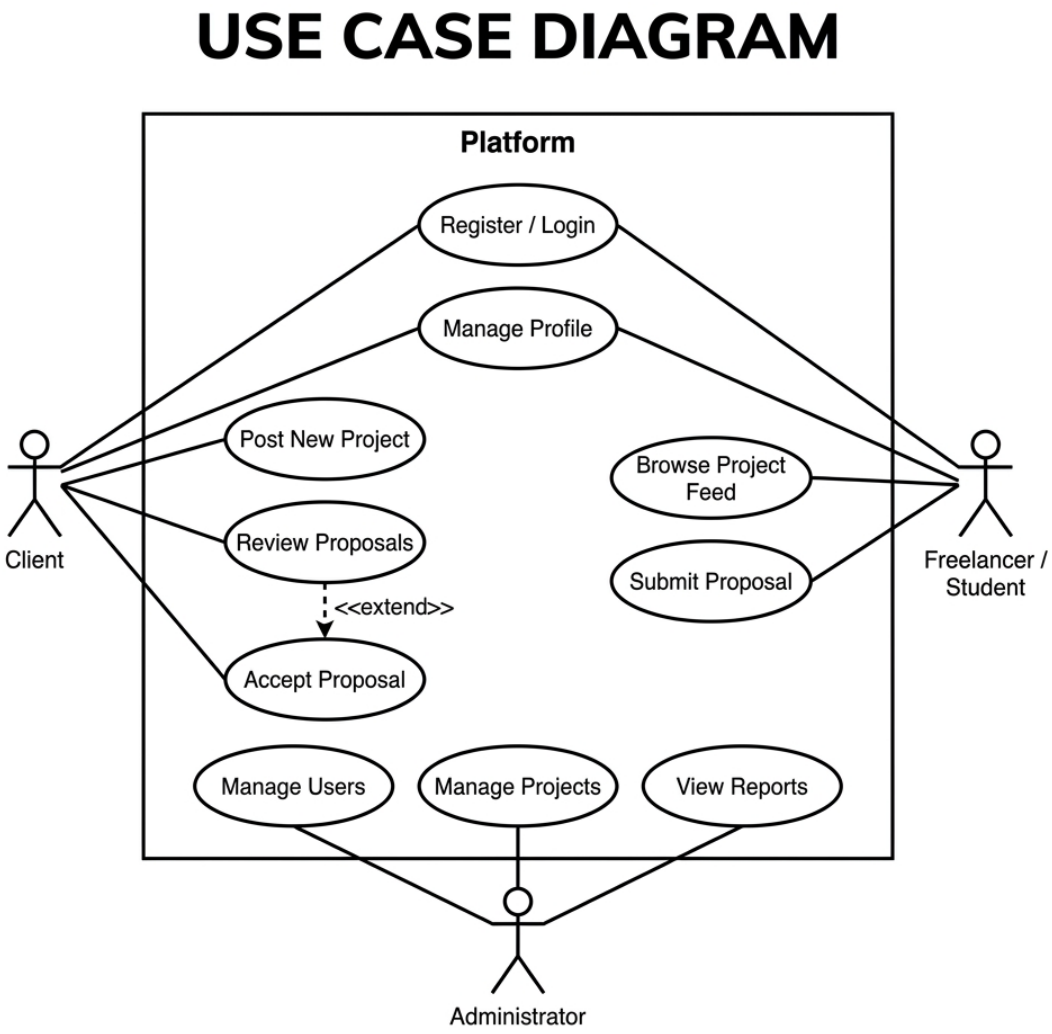
2.2. Technology Stack

Layer	Technology	Role
Frontend	Next.js 14+	SSR / SSG pages, API Routes, Server Actions
Auth	Supabase Auth	JWT sessions, OAuth, email verification
Database	PostgreSQL (Supabase)	Primary relational data store
Security	Row Level Security	Per-row DB access policies
Realtime	Supabase Realtime	Live proposal & message updates

Layer	Technology	Role
Payments	Stripe	Payment processing, escrow, invoicing
Email	Resend / SendGrid	Transactional emails & notifications
Deployment	Vercel (planned)	Edge-optimised Next.js hosting
File Storage	Supabase Storage	KYC docs, portfolio files, deliverables

Use Case Diagram

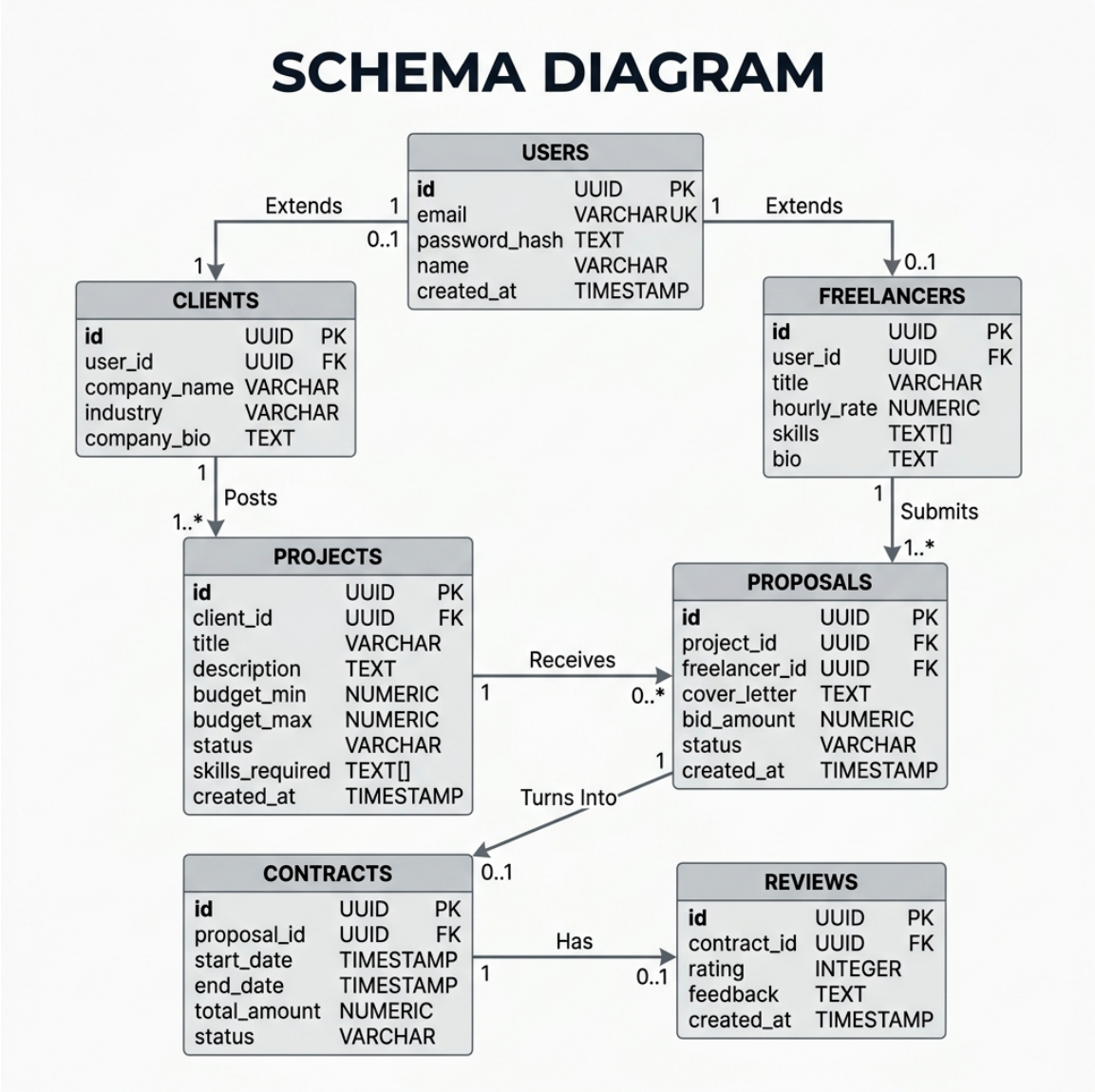
The use case diagram illustrates all interactions between the three actor types — Client, Freelancer, and Administrator — and the platform. Shared use cases (Register/Login, Manage Profile) are accessible by both Client and Freelancer roles.



Actor	Key Use Cases
Client	Post & manage jobs; Review & accept proposals; Fund escrow / milestones; Rate freelancers; File disputes
Freelancer	Build profile & portfolio; Browse & apply to jobs; Track time & submit work; Withdraw earnings; Rate clients
Administrator	Manage users & taxonomy; Moderate content; Configure fees; Resolve disputes; View analytics

Database Schema Diagram

The database is hosted on Supabase PostgreSQL and follows a normalised relational design. Seven core tables form the data backbone of the platform. All tables use UUID primary keys and enforce referential integrity through foreign key constraints and Row Level Security (RLS) policies.



Relationship	Type	Description
users → clients	1-to-1	A user account can be extended as a Client profile
users → freelancers	1-to-1	A user account can be extended as a Freelancer profile
clients → projects	1-to-many	A client may post multiple projects
freelancers → proposals	1-to-many	A freelancer may submit proposals to many projects
projects → proposals	1-to-many	A project receives multiple proposals
proposals → contracts	1-to-1	An accepted proposal generates exactly one contract
contracts → reviews	1-to-1	A completed contract may have one two-way review

1. User & Account Management

Supports dual account types, Client and Freelancer, with some platforms allowing one account to switch between roles. Registration and login are handled via email, OAuth/Google, and phone verification. Profiles capture bio, skills, portfolio, work history, hourly rate, education, and certifications, backed by identity verification (KYC) through ID upload and video verification, and a profile completeness score or strength meter. Freelancers can set an availability status (open to work, busy, etc.) and language proficiency tags, and all users get account settings covering notification preferences and privacy controls.

2. Job Posting & Discovery

Clients create job postings with a title, description, budget type (fixed or hourly), required skills, duration, and experience level, organized under a job categories/subcategories taxonomy. Freelancers find work through search and filtering by skills, budget, location, ratings, and job success score, along with saved searches and job alerts. A recommendation/matching algorithm surfaces suggested jobs and freelancers, clients can send direct job invitations to a specific freelancer, and repeat clients can save draft jobs and reusable job templates.

3. Proposals & Bidding

Freelancers submit proposals/bids with a cover letter, bid amount, estimated duration, and attachments. A connects/credits system limits spam bidding (similar to Upwork's "Connects"), and freelancers can pay to boost a proposal so it stands out. Freelancers get a proposal tracking dashboard, while clients can shortlist or archive candidates as they review submissions.

4. Contracts & Work Management

Contracts can be created as fixed-price milestones or hourly agreements, with milestone management covering funding, approval, and release of funds. Hourly contracts are backed by a time-tracking app/tool with screenshots and activity levels, plus a work diary/logged-hours review, and every contract supports task/to-do lists, amendments, extensions, and early termination. Deliverables are exchanged through file sharing, with in-platform messaging tied to the specific contract.

5. Payments & Financial

An escrow system holds funds until a milestone or piece of work is approved, with support for multiple payment methods (card, bank transfer, PayPal, wallet) and multi-currency support and conversion. Invoicing and receipts are generated automatically, and the platform applies a commission/fee structure that can be a flat rate or tiered by lifetime billing. Freelancers withdraw earnings via bank transfer, PayPal, Payoneer, or wire, with payment protection for hourly work backed by verified time tracking, dispute-triggered payment holds, region-specific tax document handling (1099s, VAT invoices), and a refund-handling workflow.

6. Communication

Real-time messaging supports text, file attachments, and emojis, with optional video/audio call integration for richer conversations. A notification system spans email, push, and in-app channels, and messaging includes read receipts and typing indicators.

7. Trust, Ratings & Reputation

A two-way review/rating system lets clients rate freelancers and vice versa, feeding into a Job Success Score/reputation algorithm. Verified badges (payment verified, identity verified, top-rated) and a portfolio showcase with endorsements build additional trust, while any dispute or complaint history factors into a user's overall trust score.

8. Dispute Resolution

Either party can start a formal dispute filing process, which moves through a mediation/arbitration workflow with admin/support intervention tools available when needed. Both sides can submit evidence such as chat logs, files, and time logs to support their case.

9. Admin & Platform Management

An admin dashboard covers user management, content moderation, and analytics, supported by a fraud/spam detection queue and tools to manage the category/skill taxonomy and fee/commission configuration. Admins moderate job posts, profiles, and messages for policy violations, review analytics/reporting such as GMV, active users, churn, and dispute rate, and operate a customer support ticketing system.

10. Additional/Value-Add Services

Additional features can include talent badges (Top Rated, Rising Talent, Expert-Vetted), skill tests and certifications, and agency accounts for teams of freelancers. AI-assisted proposal writing or job matching, a freelancer availability calendar, membership/subscription tiers (e.g., more connects at a higher plan), and a referral program round out the value-add layer.

Security Concerns :

1. Authentication & Access

Strong password policies plus multi-factor authentication (MFA); secure session management with JWT/token expiration; rate limiting on login attempts to prevent brute-force attacks; and secure OAuth token handling so client secrets are never leaked.

2. Payment Security

PCI-DSS compliance if handling cards directly, or offload to a processor such as Stripe/Adyen/Braintree. Escrow funds must be isolated from operating accounts, transactions monitored for fraud (unusual withdrawal patterns, chargebacks), and all payment webhooks validated for authenticity.

3. Data Protection

Encryption at rest and in transit (TLS everywhere); PII handling compliant with GDPR/CCPA depending on the user base; secure, access-restricted storage of KYC documents; and clear data retention and deletion policies (right to be forgotten).

4. Platform Abuse Prevention

Detection of users attempting to move payments off-platform to avoid fees (circumvention); detection of contact-info sharing in messages/proposals before a contract is signed; fake profile and fake job posting detection (bots, scams); spam bid/proposal prevention; and moderation of scraped or plagiarized portfolios.

5. Application-Level Security

Input validation and sanitization against SQL injection and XSS; malware scanning on file uploads; CSRF protection on state-changing requests; API rate limiting and abuse monitoring; strict role-based access control (admin vs. client vs. freelancer); and secure handling of sensitive time-tracking screenshots.

6. Dispute & Trust Integrity

Prevention of review manipulation (fake reviews, review bombing); complete audit logs for disputes, admin actions, and payment releases; and safeguards against chargeback fraud after work has already been delivered.

7. Infrastructure

DDoS protection at the network edge, regular penetration testing, a tested backup and disaster-recovery plan, and ongoing dependency/library vulnerability scanning.

Users and Their Privileges :

User	Role	Main Privileges
Client	Posts work and hires talent.	Register/login; post jobs; search & invite freelancers; review proposals; fund escrow/milestones; message freelancers; approve deliverables and release payment; rate freelancers; file disputes.

User	Role	Main Privileges
Freelancer	Finds work and delivers services.	Register/login; build profile and portfolio; search/browse jobs; submit proposals using connects; manage contracts; track time and log work; submit deliverables; withdraw earnings; rate clients; file disputes.
Admin	Manages overall platform operations.	Manage users, categories, and skill taxonomy; configure fees/commissions; moderate content and flagged accounts; access analytics and reporting; intervene in and resolve disputes; suspend or ban accounts; manage badges and verification status; oversee platform security and infrastructure.